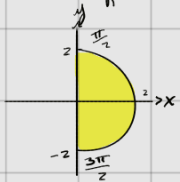


① Calcule  $\iint_R (x^2 + y^2) dx dy$

|                     |                     |
|---------------------|---------------------|
| $x = r \cos \theta$ | $x^2 + y^2 = r^2$   |
| $y = r \sin \theta$ | $dA = r dr d\theta$ |

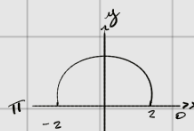


$$x^2 + y^2 = r^2 \rightarrow \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^2 r^2 r dr d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left. \frac{r^4}{4} \right|_0^2 d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4 d\theta = 4 \theta \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 4 \left( \frac{\pi}{2} - \left( -\frac{\pi}{2} \right) \right) = 4\pi$$

\* Observe que o domínio  $r$  é 2, portanto, no círculo a variação vai ser de 0 a 2

\* Poderia fazer o cálculo utilizando a simetria de  $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$ ?

②  $\iint_R \sin(x^2 + y^2) dx dy$



$$\iint_R \sin(x^2 + y^2) dx dy = \int_0^{\pi} \int_0^2 \sin(r^2) r dr d\theta = \int_0^{\pi} \left. \frac{1 - \cos(4)}{2} \right|_0^2 d\theta = \frac{\pi}{2} (1 - \cos(4)) //$$

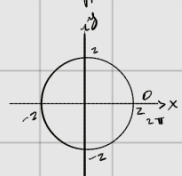
$$\int_0^2 \sin(r^2) r dr = \int_0^2 \sin(u) \frac{1}{2} du = \frac{1}{2} \int_0^2 \sin(u) du = \frac{1}{2} \left( -\cos(u) \right) \Big|_0^2 = \frac{1}{2} (-\cos(4) + 1) = \frac{1 - \cos(4)}{2}$$

$$u = r^2$$

$$\frac{du}{dr} = 2r \quad dr = \frac{du}{2r}$$

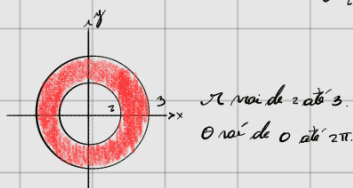
$$\int_0^{\pi} \frac{1 - \cos(4)}{2} d\theta = \int_0^{\pi} \frac{1}{2} d\theta - \int_0^{\pi} \frac{\cos(4)}{2} d\theta = \frac{1}{2} \theta \Big|_0^{\pi} - \frac{\cos(4)}{2} \theta \Big|_0^{\pi} = \frac{\pi}{2} - \frac{\cos(4) \pi}{2} = \frac{\pi}{2} (1 - \cos(4))$$

③ a)  $\iint_R \sqrt{x^2 + y^2} dA$  sendo o região, um círculo no origem e raio 2



$$\iint_R \sqrt{x^2 + y^2} dA = \int_0^{2\pi} \int_0^2 r r dr d\theta = \int_0^{2\pi} \left. \frac{r^3}{3} \right|_0^2 d\theta = \int_0^{2\pi} \frac{8}{3} d\theta = \frac{8}{3} \theta \Big|_0^{2\pi} = \frac{16\pi}{3}$$

b)  $\iint_R e^{x^2 + y^2} dA = \int_0^{2\pi} \int_2^3 e^{r^2} r dr d\theta = \int_0^{2\pi} \left. \frac{e^u - e^4}{2} \right|_2^3 d\theta = \frac{1}{2} (e^9 - e^4) \int_0^{2\pi} 1 d\theta = \frac{1}{2} (e^9 - e^4) \cdot 2\pi = \pi (e^9 - e^4)$



$$\int_2^3 e^{r^2} r dr = \int_4^9 e^u \frac{1}{2} du = \frac{1}{2} \int_4^9 e^u du = \frac{1}{2} (e^9 - e^4)$$

$$u = r^2 \quad \frac{du}{dr} = 2r \quad dr = \frac{du}{2r}$$

$$c) \iint_R \sqrt{x^2 + y^2} \, dA$$

\* como descrever a região?

$$x^2 + y^2 = 2x$$

$$x^2 + y^2 = 4x$$

+ podemos fazer a

$$|x^2 - 2x| + y^2 = 0$$

$$|x^2 - 4x| + y^2 = 0$$

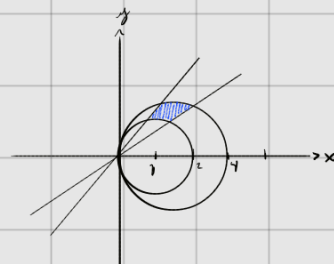
completando os quadrados.

$$|(x-1)-1| + y^2 = 0$$

$$|(x-2)-4| + y^2 = 0$$

$$(x-1)^2 + y^2 = 1$$

$$(x-2)^2 + y^2 = 4$$



$$x^2 + y^2 = 2x \rightarrow r^2 = 2r \cos \theta \rightarrow r = 2 \cos \theta$$

$$x^2 + y^2 = 4x \rightarrow r^2 = 4r \cos \theta \rightarrow r = 4 \cos \theta$$

$$* y = x \rightarrow r \cos \theta = r \sin \theta \rightarrow \theta = \frac{\pi}{4}$$

$$\begin{cases} 2 \cos \theta \leq r \leq 4 \cos \theta \\ \frac{\pi}{6} \leq \theta \leq \frac{\pi}{4} \end{cases}$$

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \int_{2 \cos \theta}^{4 \cos \theta} r^2 \, dr \, d\theta$$

$$* y = \frac{\sqrt{3}}{3}x \rightarrow r \sin \theta = \frac{\sqrt{3}}{3} r \cos \theta \rightarrow \tan \theta = \frac{\sqrt{3}}{3} \rightarrow \theta = \frac{\pi}{6}$$

$$R: \frac{7}{9} (20\sqrt{2} - 11)$$